

Apply Control Structures — Answers

Q1. Prime number check.

The following program checks whether a given number n is prime by testing divisibility from 2 up to $n/2$. If any divisor is found, the flag is set to 1, indicating the number is not prime.

```
public class PrimeCheck {
    public static void main(String[] args) {
        int n = 7, flag = 0;

        for (int i = 2; i <= n / 2; ++i) {
            if (n % i == 0) {
                flag = 1;
                break;
            }
        }

        System.out.println(flag == 0 ? "Prime" : "Not Prime");
    }
}
```

For $n = 7$, no value of i between 2 and 3 divides 7 evenly, so flag remains 0 and the program prints Prime. Adding a break once a divisor is found makes the loop stop early, avoiding unnecessary iterations once the number is known not to be prime.

Q2. Factorial.

The following program computes the factorial of n , defined as the product of all positive integers from 1 to n , using an iterative for loop.

```
public class Factorial {
    public static void main(String[] args) {
        int n = 5;
        int fact = 1;

        for (int i = 1; i <= n; i++) {
            fact *= i;
        }

        System.out.println("Factorial of " + n + " is " + fact);
    }
}
```

```
}
```

For $n = 5$, fact is updated across the iterations as 1, 1, 2, 6, 24, 120, so the final printed result is 120. The variable fact is initialized to 1 (not 0) since multiplying by 0 at the start would make every subsequent product equal to 0.

Q3. Largest of three numbers.

The following logic determines the largest of three values a, b, and c using nested selection structures.

```
public class LargestNumber {  
    public static void main(String[] args) {  
        int a = 12, b = 45, c = 30;  
  
        if (a > b && a > c) {  
            System.out.println(a);  
        } else if (b > c) {  
            System.out.println(b);  
        } else {  
            System.out.println(c);  
        }  
    }  
}
```

The logic works as follows: first, it checks whether a is greater than both b and c; if so, a is the largest. Otherwise, it checks whether b is greater than c; if so, b is the largest. If neither of the first two conditions holds, c must be the largest (or tied for largest). For $a = 12$, $b = 45$, $c = 30$, the first condition fails, the second condition ($b > c$, i.e., $45 > 30$) holds, so the program prints 45.